

Algebra PhD Exam, September 2003

Do seven of the following eleven problems. Do not turn in more than seven problems. You must show your work. Answers with no work and/or no explanations will receive no credit. State clearly any theorem you use in your proofs. In the problems, $\mathbb{Z}$, resp. $\mathbb{Q}$, $\mathbb{C}$, is the set of all integers, resp. of all rational numbers, of all complex numbers.

1. Consider the following statement $A(P)$: “If a normal subgroup H of a group G and the quotient G/H both have property P , then so does G .” Prove or disprove $A(P)$, where

- a) P is “being solvable”;
- b) P is “being nilpotent”.

2. Inside the symmetric group S_7 , consider the subgroup G generated by the permutations (1234567) and $(235)(476)$. Let

$$H = \langle a, b \mid a^7 = b^3 = 1, bab^{-1} = a^2 \rangle.$$

Show that $G \simeq H$.

3. Let G be a finite group with exactly one maximal subgroup. Prove that G is cyclic of prime power order.

4. Let F be a finite field of finite cardinality q and n be any natural number. Show that there is at least one irreducible polynomial $f \in F[x]$ of degree n .

5. Let A be a commutative ring with identity. Suppose that M and N are free A -modules with m and n generators, respectively. Prove that if $M \simeq N$, then $m = n$.

6. This problem has two parts.

- a) Give an example of a projective $\mathbb{Z}$ -module and an example of an injective $\mathbb{Z}$ -module.
- b) Let A be a finite abelian group considered as a $\mathbb{Z}$ -module and let I be an injective $\mathbb{Z}$ -module. Compute $A \otimes_{\mathbb{Z}} I$.

7. Let $\alpha \in \mathbb{C}$ be such that $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 2003$. Let $E/\mathbb{Q}$ be a normal closure of $\mathbb{Q}(\alpha)/\mathbb{Q}$. Prove that $[E : \mathbb{Q}]$ divides $2003!$.

8. Let $\xi \in \mathbb{C}$ be a primitive 77^{th} root of unity in $\mathbb{C}$, and let $F = \mathbb{Q}(\xi)$.

- a) Briefly explain why $F/\mathbb{Q}$ is a Galois extension, and describe the structure of the Galois group $\text{Aut}_{\mathbb{Q}}(F)$.
- b) Find the number of subfields of F that are quadratic extensions of $\mathbb{Q}$.

9. State and prove the Hilbert Basis Theorem.

10. Let D be a Dedekind domain and F be the group of fractional ideals of D . Prove that any nontrivial element in F has infinite order.

11. Let R be a commutative ring with identity such that every element $x \in R$ satisfies $x^{n(x)} = x$ for some integer $n(x) \geq 1$ depending on x . Prove that the Jacobson radical of R is 0.